

Bhanu Sai Prakash Chikkavarapu

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About Me

Mechanical Engineer with 5+ years of experience in new product development, performing engineering calculations and FEA to support design decisions, prototyping, and validation. Strong background in materials, DFMA, and automated test setup development with sensor/DAQ integration for design verification and performance testing. Skilled in technical drawings, engineering support, and troubleshooting.

Professional Experience

Graduate Research Assistant – Northern Illinois University Aug 2024 – Present

- Designed a compact and modular filtration test setup, improving leak tightness and measurement accuracy by 15%.
- Integrated with sensors, DAQ hardware, and control components for performance evaluation and validation.
- Developed staged filtration concepts and optimized internal flow paths to improve efficiency and reliability.
- Created fixture designs, and modular components that support ongoing R&D and iterative product development.

Design Engineering Intern – Hydraulic Technologies May 2026 – Aug 2026

- Development of a highly engineered two-stage hi-low radial piston pump using FEA-driven design optimization.
- Performed competitors product teardown analysis to compare design intent and evaluate component life ratings.
- Conducted analytical, FEA, and experimental buckling studies of high-tonnage aluminum cylinders.
- Designed and fabricated test fixtures to support multiple cylinder configurations for cycle and side-load testing.

Mechanical Engineer Intern – APSCO, Inc. Jun 2025 – Aug 2025

- Designed and fabricated a compact multi-stage inlet air filtration system for mobile blower applications.
- Reduced overall filter system size by 27% and increased service life by 15× through staged filtration.
- Designed fixtures and prototype hardware enabling fast iteration and repeatable testing during development.
- Collaborated with cross-functional teams to refine manufacturability, serviceability, and system layout.

Mechanical Design Engineer – Design Cube Engineering Solutions Pvt. Ltd. Jul 2021 – Aug 2024

- Designed oil reservoirs with cooling and filtration systems for mobile applications, while ensuring structural integrity.
- Created detailed 2D/3D models and drawings with GD&T, and BOMs using CAD softwares.
- Applied DFMA principles to existing products, reducing part count by 15–20% and shortened assembly time by 18%.
- Performed FEA to optimize material usage and ensure product reliability, achieving 10–17% cost and weight savings.
- Worked with manufacturing and sales teams to provide technical support and resolve engineering issues.

Skills

Engineering	CAD & Design	Manufacturing	Analysis	Software
NPD/EPD	3D/2D CAD	CNC Machining	Finite Element Analysis	Solidworks/Creo
DFM/DFA	GD&T (ASME Y14.5)	Sheet Metal	Tolerance Stack-Up	Ansys
Material selection	Fixtures/Test Setup	Additive Manufacturing	System Troubleshooting	MATLAB

Education

Northern Illinois University, IL
Master of Science in Mechanical Engineering Aug 2024 – [Expected] Dec 2026

VR Siddhartha Engineering College, India
Bachelor of Technology in Mechanical Engineering 2017 – 2021